

LCA at Tallgrass: One-Page Brief

Oct 2, 2026 · Andrew Gibson

How an LCA works

An LCA counts every greenhouse gas emitted and removed over a product's life. For Trailblazer, it sets the carbon intensity (CI) that scales 45Z and clean fuel credits, and the removal-credit yield. Four phases (ISO 14040/44):

- 1. **Goal and scope:** what unit, which boundaries (corn field to Wyoming storage).
- 2. **Inventory:** every input and output, from fertilizer to pipeline power.
- 3. **Impact:** convert every flow to CO2-equivalent and total it.
- 4. **Interpretation:** test what drives the answer (land use, grid power, leakage).

The programs Trailblazer can use

Program	Pays per	Land-use charge (iLUC)	Value	Market and outlook
45Z	Gallon, scaled by fuel CI	Excluded for fuel made after 2025	~\$195–260 per tonne captured (est.)	\$12.5B (FY2026, all fuels); ends Dec 31, 2029
45Q	Tonne stored	Not counted	\$85 per tonne	\$1.0B in FY2026, rising to \$5.9B by FY2035
Clean fuel (CA, OR, WA)	Tonne CO2e below the CI standard	Included in CA	\$80–87 per credit (Sep 30, 2026)	CA ~\$190 (2019), ~\$60 (2025); needs a CCS pathway
Removal credits (Puro)	Net tonne removed	Charged; about half of each captured tonne (est.)	\$100–200 per credit (est.)	30.4 Mt contracted in 2025, ~90% by Microsoft

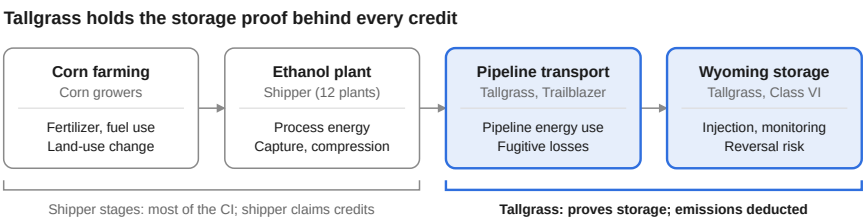
A plant claims 45Z or 45Q in a year, never both. Puro bars reusing a removal in other environmental products, so a tonne earns clean fuel or removal credits, not both. iLUC: carbon released when corn for fuel pushes farming onto new land.

Cap-and-trade: a market to shape

Cap-and-trade is spreading. California's program (linked with Quebec) runs to 2045; Washington signed to join in June 2026, with a joint market expected in 2027. None of these, nor RGGI, credits CO2 stored in Wyoming.

- **Most realistic:** California's SB 840 requires CARB to consider carbon removal offset protocols by 2029. Push for one that accepts out-of-state Class VI storage.
- **Long shot:** a Wyoming or Nebraska program. Wyoming lawmakers tried to repeal even a CCS study mandate in 2026.

Where Tallgrass sits



How Tallgrass applies it

- **Own the storage-stage LCA:** one audited data package that all 12 shippers can cite, for a fee or revenue share.
- **Grade shippers' tonnes by full LCA:** premium tonnes to strict buyers like Microsoft, the rest to spot.
- **Screen new CO2 sources:** coal earns only 45Q (\$85/t) against capture costs of ~\$60–100/t (industry estimates); it fits only as take-or-pay volume.

Sources: Treasury Tax Expenditures FY2027; FuSE prices (Sep 30, 2026); CDR.fyi; Puro GSC 2024 v.5; ICAP; WA Ecology. ~ or est. = estimate.